

Wrong rendering order of particles, hitboxes, clouds, transparent blocks, breaking animations and various other transparent textures

MC-9553 Rendering

Status	Resolved / Fixed
Affected	Snapshot 13w06a, Snapshot 13w07a, Snapshot 13w09a, Snapshot 13w09b, Snapshot 13w09c, Snapshot 13w10a (+306 more)
Fixed in	19w39a
Reported	2013-02-10

STEPS TO REPRODUCE

1. Launch Minecraft 18w22a.
2. Create a new Creative world with default settings.
3. Place water, ice, and stained glass in the world.
4. Break stained glass in front of the water and use `/weather rain`` to produce particles.
5. Move above `y=128` so that the clouds become transparent.
6. Enable the hitbox display using the F3 debug screen and view the water, ice, and stained glass.
7. Run `/summon creeper ~ ~1 ~ {powered:1b,ActiveEffects:[{Id:14,Duration:2000000}]}``.
8. Switch to third-person view and look at the ice.
9. Observe that particles, hitboxes, transparent clouds, the charged creeper energy field, and ice render in the wrong order relative to water or other transparent blocks; some particles and transparent textures render behind or in place of the transparent blocks, hitboxes may be obscured, clouds render in front, and ice renders incorrectly in third person.

OBSERVED BEHAVIOUR

Actual behavior

Various translucent render layers are drawn in an inconsistent order, causing some objects to appear behind, in front of, or to replace other transparent objects depending on the viewpoint and game version.

Observed cases include:

- End rod, potion, firework, ink, rain, and snow particles can render behind water, ice, stained glass, transparent clouds, hitboxes, or block-breaking animations, making them partially or almost completely invisible. In older versions, this affected a broader range of particle types.
- Block-breaking animations and other non-transparent particles may render incorrectly relative to transparent blocks. In snapshot 18w22a, for example, breaking a block in front of water can render the effect in the same depth layer as the water rather than clearly in front of it.
- Hitboxes have changed ordering between versions. In 13w41a they could render in front of water, ice, and stained glass, including showing normally hidden portions through those blocks. In later versions, including 1.7 and snapshot 18w22a, hitboxes can instead render behind transparent blocks. Snapshot 18w22a fixed hitboxes rendering underwater and through water, but they remained difficult to see through ice and colored glass.
- Clouds become translucent above `Y=128` and render in front of transparent blocks and hitboxes, including water and glass. This ordering is particularly noticeable when translucent clouds overlap these objects.
- The energy field of an invisible charged creeper renders behind water and translucent clouds but in front of hitboxes. Other particles and transparent blocks do not consistently share this ordering.
- The flash from a firework explosion renders over water and translucent clouds, causing those layers to disappear while the flash is visible.
- The outer translucent portions of beacon beams and phantom wings, as well as the translucent text overlay of named map markers near a map's edge, render behind water, translucent clouds, and other transparent blocks, causing them to be obscured.
- Ice blocks have incorrect rendering in third-person view and can also render incorrectly at far distances.

These ordering problems remain visible in snapshot 18w22a for translucent particles, charged-creeper auras, phantom wings, the outer beacon-beam aura, firework particles, and clouds above the cloud height.

EXPECTED BEHAVIOUR

Particles, hitboxes, clouds, transparent blocks, breaking animations, beacon beams, entity effects, firework flashes, map markers, phantom wings, and other translucent textures should render in the correct depth order relative to one another. They should remain visible when appropriate without appearing through nearer surfaces, obscuring transparent blocks, or causing water, clouds, ice, or other translucent elements to disappear or render incorrectly.

ENVIRONMENT

Minecraft Java Edition Versions: 13w06a, 13w07a, 13w09a, 13w09b, 13w09c, 13w10a, 13w10b, 1.5, 13w11a, 1.5.1, 13w16a, 13w16b, 1.5.2, 13w17a, 13w18a, 13w18b, 13w18c, 13w19a, 13w21a, 13w21b, 13w22a, 13w23a, 13w23b, 13w24a, 13w24b, 13w25a, 13w25b, 13w25c, 13w26a, 1.6, 1.6.1, 1.6.2, 1.6.3, 1.6.4, 13w36a, 13w36b, 13w37a, 13w37b, 13w38a, 13w38b, 13w38c, 13w39a, 13w39b, 13w41a, 13w41b, 13w42a, 13w42b, 13w43a, 1.7, 1.7.1, 1.7.2, 13w47a, 13w47b, 13w47c, 13w47d, 13w47e, 13w48a, 13w48b, 13w49a, 1.7.3, 1.7.4, 14w02a, 14w02b, 14w02c, 14w03a, 14w03b, 14w04a, 14w04b, 14w05a, 14w05b, 14w06a, 14w06b, 14w07a, 14w08a, 1.7.5, 14w10b, 14w10c, 14w11b, 1.7.9, 14w17a, 14w18a, 14w20a, 14w20b, 14w21a, 14w21b, 14w25b, 1.7.10, 14w28a, 14w28b, 14w29a, 14w29b, 14w30b, 14w30c, 14w31a, 14w32a, 14w32b, 14w32c, 14w32d, 14w33c, 14w34a, 14w34b, 14w34c, 14w34d, 1.8-pre1, 1.8-pre2, 1.8, 1.8.1, 1.8.3, 1.8.4, 1.8.8, 15w31c, 15w32a, 15w32b, 15w32c, 15w33b, 15w33c, 15w34d, 15w35b, 15w35e, 15w36b, 15w36c, 15w36d, 15w39c, 15w40b, 15w41b, 15w42a, 15w43a, 15w43c, 15w44a, 15w44b, 15w45a, 15w46a, 15w47a, 15w47c, 15w50a, 15w51b, 16w05b, 16w06a, 16w07a, 16w07b, 1.9 Pre-Release 1, 1.9 Pre-Release 2, 1.9 Pre-Release 3, 1.9 Pre-Release 4, 1.9, 1.9.1 Pre-Release 1, 1.9.1 Pre-Release 2, 1.9.1 Pre-Release 3, 1.9.2, 1.9.3 Pre-Release 1, 1.9.3 Pre-Release 3, 1.9.4, 16w20a, 16w21a, 16w21b, 1.10 Pre-Release 1, 1.10 Pre-Release 2, 1.10, 1.10.1, 1.10.2, 16w32b, 16w33a, 16w35a, 16w36a, 16w39b, 16w39c, 16w40a, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 16w50a, 1.11.2, 17w06a, 17w13b, 17w16b, 17w17b, 17w18a, 17w18b, 1.12 Pre-Release 1, 1.12 Pre-Release 2, 1.12 Pre-Release 3, 1.12 Pre-Release 4, 1.12 Pre-Release 5, 1.12 Pre-Release 6, 1.12 Pre-Release 7, 1.12, 17w31a, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2 Pre-Release 1, 1.12.2 Pre-Release 2, 1.12.2, 17w43a, 17w43b, 17w45b, 17w46a, 17w47a, 17w47b, 17w48a, 18w03b, 18w05a, 18w06a, 18w07a, 18w07b, 18w07c, 18w08a, 18w08b, 18w09a, 18w10a, 18w10b, 18w10c, 18w10d, 18w11a, 18w14a, 18w14b, 18w15a, 18w16a, 18w19a, 18w19b, 18w20a, 18w20b, 18w20c, 18w21a, 18w21b, 18w22a, 18w22b, 18w22c, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre5, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13-pre9, 1.13-pre10, 1.13, 18w30a, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1-pre2, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w43a, 18w43b, 18w43c, 18w44a, 18w45a, 18w46a, 18w47a, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03a, 19w03b, 19w03c, 19w04a, 19w04b, 19w05a, 19w06a, 19w07a, 19w08a, 19w08b, 19w09a, 19w11a, 19w11b, 19w12b, 19w13a, 19w13b, 19w14a, 19w14b, 1.14 Pre-Release 1, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14 Pre-Release 5, 1.14, 1.14.1 Pre-Release 1, 1.14.1, 1.14.2 Pre-Release 1, 1.14.2 Pre-Release 2, 1.14.2 Pre-Release 3, 1.14.2 Pre-Release 4, 1.14.2, 1.14.3 Pre-Release 1, 1.14.3 Pre-Release 2, 1.14.3 Pre-Release 3, 1.14.3 Pre-Release 4, 1.14.3, 1.14.4 Pre-Release 1, 1.14.4 Pre-Release 2, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.14.4 Pre-Release 5, 1.14.4 Pre-Release 6, 1.14.4 Pre-Release 7, 1.14.4, 19w34a, 19w35a, 19w36a, 19w37a, 19w38b, 19w38a (includes snapshots)

ORIGINAL DESCRIPTION

Mod Notice

Please do not attach any more images to this ticket.

For any new instances of this issue occurring (or similar), please perform a search on the bug tracker for reports, else create a new ticket for the specific issue being observed. To better address issues moving forward, individual problems will be tracked separately, instead of a "global" report such as this.

Affected cases

- Some particles render behind transparent blocks, hitboxes, breaking animations and transparent clouds, often making them barely visible. Particles known to have this issue: End rod, potion, firework, ink, rain and snow particles. In older versions, this was a general issue with all types of particles, some have been fixed since then.
- As of 13w41a, hitboxes render in front of transparent blocks - it simply looks weird when you can see the hitbox of blocks through the water surface (see water_hitbox.jpg) or the rear (normally less visible) parts of the hitbox through a transparent block like ice or stained glass. As of 1.7, the behavior changed: Hitboxes now render behind

transparent blocks!

- Transparent clouds 1 always render in front of transparent blocks and hitboxes (looks pretty weird, see clouds_glass.jpg).

- The energy field around invisible charged creepers renders behind water and transparent clouds, but in front of hitboxes. Particles and other transparent blocks show no rendering order problems in this case. You can spawn invisible charged creepers with the following command:

```
/summon creeper ~ ~1 ~ {powered:1b,ActiveEffects:[{Id:14,Duration:2000000}]}
```

- The firework explosion light flash causes water and transparent clouds to always be rendered behind it, making them disappear.

- The outer, transparent part of the beacon beam, the transparent text overlay of named map markers (close to a map's edge) as well as the outer transparent part of Phantom wings render behind water, transparent clouds and transparent blocks, making them disappear.

- Ice blocks render incorrectly in third person

Notes

1 "Transparent clouds" means: Clouds turn transparent when you move above y=128 (

MC-997

Resolved

). Before Jeb reverted the fix of

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-9553>